SD2 Lab Project Report

International Islamic University Chittagong (IIUC)

Department of Computer Science & Engineering

Project Title:

Interactive Developer Portfolio Application Using Flutter and GetX

Submitted By

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Course: SD2 Lab

Project Type: Flutter Portfolio Application Development

1. Introduction

This project aims to develop an interactive and responsive developer portfolio application using Flutter. It presents personal information, skills, academic background, and completed projects in an organized way. Modern UI design and animations are used to improve the user experience. GetX architecture is implemented to separate the user interface from the application logic. The project reflects the practical knowledge gained from the SD2 Lab course, including Flutter development, responsive design, animation, and state management.

2. Project Features

The developed Flutter portfolio application contains the following sections:

Home Section

The Home section introduces the developer with:

- Profile image
- Name and professional role
- Short introduction
- Action buttons

About Me Section

This section contains:

- Personal biography
- Educational background
- Professional interests
- Developer information

Skills Section

Technical skills are displayed using interactive animated cards.

Examples:

- Flutter
- Dart
- Firebase
- GetX
- React
- Next.js
- Node.js

Projects Section

The project section displays previous works with:

- Project title
- Description
- Technologies used
- Project image
- Live project link
- GitHub repository link

Contact Section

The contact section provides interactive links for:

- Email
- Phone
- Location
- GitHub
- LinkedIn

3. Animation Implementation

One of the main requirements of this project was implementing size and color transition animations in at least three different UI components.

3.1 Hero Button Animation

The Hero section contains an animated button using AnimatedContainer.

Implemented transitions:

- Button width change
- Button height change
- Background color change
- Shadow effect

This creates a smooth interactive experience when the user interacts with the button.

3.2 Profile Card Animation

The profile/about card contains interactive size and color transitions.

Implemented animations:

- Card expansion
- Background color transition
- Border radius change
- Profile image size transition

When the user clicks the profile card, the component changes smoothly.

3.3 Skill Card Animation

Skill cards also contain animation effects.

Implemented transitions:

- Card size change
- Background color change
- Highlight effect

These animations improve the presentation of technical skills.

3.4 Additional Animations

Additional animations were added to:

- Project cards
- Contact cards
- Hero section elements

These effects improve the overall user experience.

4. GetX Architecture Implementation

GetX was used as the state management solution for this project.

The project follows a structured architecture:

- **Controllers:** Handles application logic and state changes.
- **Views:** Contains the main application screens.
- **Widgets:** Contains reusable UI components.
- **Models:** Represents structured application data.

Using GetX helped to separate UI code from business logic and made the application more organized and maintainable.

Reactive variables and Obx widgets were used to update UI components automatically based on state changes.

Controller

The controller manages application logic and UI states.

It handles:

- Animation states

- User interactions
- Component updates

Reactive State Management

GetX reactive programming was implemented using:

- GetxController
- Observable variables (.obs)
- Obx widgets

These features allow the UI to update automatically when the application state changes.

Dependency Injection and Routing

GetX concepts such as dependency injection, bindings, and routing help maintain a scalable and organized application structure.

5. Reflection of SD2 Lab Learning

This project reflects the practical knowledge gained throughout the SD2 Lab course.

During the lab, I learned the fundamentals of Flutter and Dart application development and how to build user interfaces using different widgets such as Scaffold, AppBar, Container, Row, Column, Stack, Text, and buttons. I also learned the differences between StatelessWidget and StatefulWidget and how state changes can update the user interface.

The course helped me understand responsive UI design and how to create layouts that work properly on different screen sizes. I also learned Flutter animation techniques, including size, position, shape, and color transitions using animation widgets such as AnimatedContainer. These concepts were directly applied in this project to create interactive UI components.

Another important learning outcome was GetX state management. I learned how to use GetxController, reactive variables, Obx widgets, dependency injection, bindings, and routing. These concepts helped me separate application logic from UI components and maintain a clean project structure.

The SD2 Lab also introduced API integration, asynchronous programming, and JSON data handling concepts. These topics improved my understanding of how Flutter applications communicate with external services and manage dynamic data. Additionally, I learned professional software development practices including creating Git repositories, using Git add, commit, and push commands, maintaining version history, publishing projects on GitHub, and preparing project documentation using README files.

Overall, this project combines the major concepts learned in SD2 Lab, including Flutter development, responsive design, animation, GetX architecture, clean coding practices, and GitHub-based project management.

6. Conclusion

The Interactive Developer Portfolio Application successfully demonstrates the implementation of Flutter, GetX architecture, responsive UI design, and animated UI components. Through this project, I was able to apply the concepts learned in SD2 Lab into a complete working application. The development process improved my understanding of Flutter widgets, state management, animation techniques, clean architecture, and professional project documentation.